Andrew Olvera

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RESEARCH INTERESTS

Agile, high-speed robotics in complex and GNSS-denied environments. Integrating robot learning, optimal control, computer vision, and learned world models for robust perception, planning, and coordination across multi-platform robot teams.

EDUCATION

York College of Pennsylvania Aug 2023 – May 2026
B.S. in Computer Science, York, PA

- Coursework: Data Structures, Algorithms, Linear Algebra, Calculus, Probability & Statistics, Computer Vision, Computer Graphics

Cabrini University Aug 2021 – May 2023
Undergraduate coursework, Radnor, PA; transferred to York College

RESEARCH EXPERIENCE

Starscream 2026 – Present
Independent research; manuscript in preparation

- Pretrained an action-conditioned physical world model and learned a continuous 64D expert-motion interface with a current-state feedback controller from successful MPCC trajectories.
- Aligned a world-model-initialized Transformer and learned MLP projector through JEPA-style stopped-target regression and expert-initialized physical rollouts on the system's own achieved states.
- Refined the observation-only planner with racing PPO, a gate-start curriculum, dynamic course sampling, and qualified online variants while keeping the low-level controller frozen.
- Recorded selected clean flights across learned Swift, A2RL, UTT, and derived graft layouts with one shared policy. Current preview uses nominal simulated dynamics, perfect state and gate map, and 130 Hz control; it is not an aggregate generalization benchmark.
- Integrated OpenVINS stereo visual-inertial odometry with rendered camera images and simulated IMU, including timestamp alignment and closed-loop validation in simulation.
- Developed C++ indoor exploration and inspection software using depth occupancy maps, viewpoint selection, and roadmap-based planning with collision-checked flight corridors.

Tells 2026 – Present
Independent research; manuscript in preparation

- Built action-conditioned latent world models and state/action tokenizers for MicroRTS, with complete-state and fog-visible representations.
- Distilled a 16-code opponent belief from fog-of-war history using privileged training targets; improved return prediction, while downstream policy benefits remained inconclusive.
- Trained PMPO actor-critic policies with eight-step imagined rollouts in a frozen learned world, alongside real-engine data collection; compared belief-conditioned and separately trained no-belief policies.

Reward Design in Small-Model RLVR 2025 – 2026
Independent research; site report

- Studied RLVR on Qwen3 0.6B (GSM8K), comparing GRPO and DAPO across group sizes, then binary, structured, and rank-based rewards on a fixed baseline.
- Exploratory single-seed runs reached roughly 69–72% accuracy from a 68% SFT baseline; tracked reward variation as binary rewards saturated.
- Analyzed token-level advantages, dynamic clipping, and reward structure to study when the optimizer retains a useful learning signal.

MANUSCRIPTS IN PREPARATION

A. Olvera. *Starscream.* In preparation, 2026.
A. Olvera. *Tells.* In preparation, 2026.

HONORS

- Dean's List, York College of Pennsylvania – Fall 2025, Spring 2026

INDUSTRY EXPERIENCE

Sapio Sciences May 2025 – Present
Application Engineer Intern, York, PA

- Develop customer-specific software solutions and build foundational features for the out-of-the-box product.
- Coordinate customer implementations across teams, communicating requirements and progress throughout delivery.

Dataforma May 2025 – Jan 2026
Software Engineer Intern, Web Development, York, PA

- Refactored legacy JSP features onto a new stack with a Vue.js frontend and Kotlin backend.
- Worked with development, testing, design, and QA teams to implement features, debug defects, and validate releases.

TECHNICAL SKILLS

ML & RL: PyTorch, Hugging Face Transformers, TRL, vLLM; imitation learning, policy gradients, model-based RL
Robotics & Control: MPC/MPCC (acados), stereo VIO (OpenVINS), occupancy mapping, motion planning; ROS, Flightmare, Isaac Sim, MATLAB/Simulink
Languages: Python, C++, C, Java, Kotlin, JavaScript, SQL
Tools: Linux, Git, Docker, Bash, Weights & Biases, FreeCAD